

LPAR2 (Q9HBW0) — GO:0007189

"adenylate cyclase-activating GPCR signaling pathway": Focused Curation Review

Focus type: function_assignment · **Hypothesis slug:** conditional-adenylate-cyclase-activation
Gene: LPAR2 (human), UniProt Q9HBW0 · **Paralog cross-checked:** LPAR3 / Q9UBY5 (EDG7)
Iteration: 3 (final) · **Date:** 2026-09-21

Executive Judgment

Verdict: Weakly / partially supported — the activating annotation is a phylogenetic over-propagation (from melanocortin receptors) whose only direct experimental anchor for LPAR2 is a conditional, heterologous in-vitro effect; it should be treated as non-core (context-qualified), not as a primary human function. Native mammalian physiology points the opposite direction (cAMP-lowering / Gi, plus Gq and G12/13).

Three lines of evidence, weighed together:

1. **The current annotation's basis is melanocortin receptors, not LPA data (decisive, new in Iteration 2).** QuickGO shows the single GO:0007189 annotation on Q9HBW0 is IBA (GO_REF:0000033, PANTHER node **PTN002733616**). Its experimental withFrom source set resolves to **MC1R, MC2R, MC3R, MC4R, MC5R** (all Gs-coupled, cAMP-activating melanocortin receptors) **plus GPR3** (a constitutively Gs-active orphan). **No LPA receptor is among the sources.** LPAR2 simply co-resides with melanocortins in PANTHER family **PTHR22750** (the functionally heterogeneous "MECA" δ-branch: Melanocortin/EDG-LPA-S1P/Cannabinoid/Adenosine). This is a textbook paralog over-annotation: a Gs/cAMP-up function experimentally proven in melanocortin receptors is inherited by LPA receptors whose real coupling is different.

2. **The only direct LPAR2 pro-cAMP datum is conditional and heterologous.** Bando et al. 1999

(P 10488122):

in EDG4/LPAR2-expressing **Sf9 insect cells**, LPA **enhanced forskolin-stimulated cAMP**, absent in EDG2/LPAR1, and **PTX-insensitive; LPA alone was ineffective** (required 5 μ M forskolin + 0.5 mM IBMX priming). This is real direct evidence but reflects **potentiation of an already-activated cyclase** — consistent with a conditional Gq/PKC $\rightarrow$ adenylyl-cyclase-type-II mechanism (the "cyclase-II/PKC" idea raised for the paralog by Im 2000,

P 10727522)

— not canonical Gs activation. Bando's own abstract frames LPA action as "*modulation of adenylyl cyclase.*"

3. **Native mammalian evidence shows Gi-mediated cAMP LOWERING and Gq/G12 (strengthened in Iteration 3).** The seed-cited

P 16203867

(Li 2005, J Exp Med) — retrieved via Europe PMC — states directly that apical intestinal **"LPA inhibited CFTR-dependent iodide efflux through LPA2-mediated Gi pathway,"** reducing cholera-toxin-induced secretion. Concordantly, disrupting LPA2-NHERF2 *raises* cAMP near CFTR (i.e., intact LPA2 lowers it;

P 21299497);

LPA **inhibits** forskolin-stimulated HCO_3^- secretion NHERF2-dependently

(P 19221439);

LPA2 couples to **Gaq** (NHERF-2) and **Ga12** (MAGI-3) in colon cancer

(P 21134377)

and to **Gi** (PTX-sensitive) in dendritic cells

(P 16769764).

None report Gs/cAMP activation.

4. **Both paralogs share the identical melanocortin-derived IBA (Iteration 3).** QuickGO confirms **LPAR3 (Q9UBY5)** carries GO:0007189 with the *same* IBA code, *same* GO_REF:0000033, *same* PANTHER node PTN002733616, and the *same* `withFrom` set (MC1R–MC5R + GPR3) as LPAR2 — i.e., uniform family-level propagation, not paralog-specific experimental support, exactly as the seed's tree analysis implied.

Answer to the seed's explicit question ("supported, merely non-core, or unresolved?"): **the activating annotation is real as a conditional in-vitro capacity but non-core; as currently coded (IBA from melanocortin receptors) it is over-propagated.** Do not silently rename "activating" to "modulating" — but do (a) re-ground it on the Bandoh IDA with a context qualifier, and (b) add the missing inhibitory (Gi) annotation so the record reflects dominant physiology. The direction is conditional, but the *basis* of the present annotation is a paralog artifact.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
P 10488122 (Bandoh 1999)	Direct assay (heterologous)	Supports (conditional)	LPAR2 activates adenylate cyclase	LPA raised forskolin-stimulated cAMP in EDG4/EDG7-Sf9 (not EDG2); PTX-insensitive ; LPA alone ineffective	Sf9 insect cells, +5 μ M forskolin/0.5 mM IBMX	Moderate; heterologous; forskolin-primed; potential not confirmed; Gs
QuickGO/UniProt Q9HBW0 IBA (GO_REF:0000033, PTN002733616)	Structural/evolutionary (annotation provenance)	Qualifies/Competing (over-propagation)	Basis of the GO:0007189 IBA on LPAR2	IBA <code>withFrom</code> = MC1R/MC2R/MC3R/MC4R/MC5R + GPR3 (Gs cAMP-activating); no LPA receptor among sources; LPAR3/Q9UBY5 shares identical IBA/withFrom	PANTHER PTHR22750 (MECA family); queried 2026-09-21	High; database read; annotation inheritance from melanocytes
P 16203867 (Li 2005, J Exp Med)	Direct assay + mouse phenotype (native intestine)	Competing (Gi / AC-inhibition)	LPAR2 cAMP-relevant coupling natively	"LPA inhibited CFTR-dependent iodide efflux through LPA2-mediated Gi pathway" ; reduced CTX-induced fluid	Human/mouse intestinal epithelium, apical CFTR complex	High; direct; native (cAMP- lowering coupling opposite); GO:000

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence and limitations
				secretion; LPA2-peptide reversible		
P 21299497 (Zhang 2011)	Perturbation (native epithelia)	Competing (cAMP-lowering)	LPAR2 raises vs lowers cAMP	Disrupting LPA2-NHERF2 elevates cAMP near CFTR → intact LPA2 lowers cAMP	Human airway epithelia	Moderate; high; open direction GO:000
P 19221439 (Singh 2009)	Mutant phenotype (mouse)	Competing (Gi-type)	LPAR2 inhibits cAMP-dependent secretion	LPA inhibited forskolin-stimulated HCO ₃ ⁻ secretion; lost in Nherf2 ^{-/-} ; "NHERF2 confers inhibitory signals"	Mouse duodenum in vivo	Moderate; high; no cAMP-lowering
P 21134377 (Lee 2011)	Interaction (native)	Competing (Gq/G12)	LPAR2 G-protein coupling	LPA2 couples to Gαq (NHERF-2) and Gα12 (MAGI-3); no Gs	Human colon cancer (HCT116/SW480)	Moderate; high; G core
P 10727522 (Im 2000)	Direct assay (mammalian)	Qualifies (paralog LPAR3)	LPAR3 cAMP/G coupling	EDG7-RH7777 show no inhibition of forskolin cAMP; HEK293T+Gi2α → GTPγS (Gi, EC ₅₀ 195 nM)	RH7777, HEK293T	High for LPAR3; LPAR2

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
P 10729222 (Contos & Chun 2000)	Sequence/genomic	Qualifies (construct)	Early EDG4 construct integrity	Ovarian-tumor EDG4 cDNA has a G-deletion frameshift near 3' end → altered C-terminus	Human/mouse genomic vs cDNA	High; outcome construction depends
P 16769764 (Oz-Arslan 2006)	Native-cell pharmacology	Competing (Gi)	Native LPAR2 coupling	PTX + PKC/Rho inhibitors block LPL-induced ERK/cytokines in LPA2 ⁺ DCs	Human dendritic cells	Moderate; Gq/Rho dominant
P 16504475 (Kim 2006)	Genetic/pharmacology	Competing (Gq)	LPA2 couples to Gq	Gq-inhibitor peptide attenuates LPA proliferation/ERK/Akt in LPA1/LPA2 VSMC	Mouse/rat VSMC	Moderate
P 16904289 (Zhang 2007)	Interaction	Competing (G12/13-Rho)	LPA2 C-terminal PDZ routing	LPA2 C-terminus binds MAGI-3/NHERF2/PDZ-RhoGEF; regulates ERK/RhoA	SW480 colon cancer	Moderate

Provenance artifact: [provenance/LPAR2_G00007189_evidence_matrix.csv](#) (machine-readable). UniProt GO/sequence and QuickGO annotation provenance pulled programmatically via UniProt REST + EBI QuickGO REST (code executed in-session).

GO Curation Implications

- **Term class:** GO:0007189 is a **Biological Process (BP)** term.
- **Lead (requires curator verification): Do not keep GO:0007189 as an unqualified IBA.**
Two coherent options:
- **Re-ground and qualify:** replace the melanocortin-derived IBA with an **experimental annotation from**

P 10488122

(IDA in a heterologous system), explicitly qualified as a **forskolin-dependent, PTX-insensitive potentiation** (i.e., conditional/non-core). This preserves the genuine capacity without implying canonical Gs coupling.

- **Down-weight/flag as over-propagated:** because the IBA `withFrom` set is entirely melanocortin/GPR3 (Gs) receptors with **no LPA-receptor source**, a curator could reasonably treat the IBA as a phylogenetic artifact of the MECA family and remove or NOT-qualify it, retaining only the qualified experimental annotation.
- **Complementary term to ADD: GO:0007193** ("adenylate cyclase-*inhibiting* GPCR signaling pathway") is currently **absent** from Q9HBW0 despite strong native-tissue support

P 21299497

P 19221439

P 16769764

Adding it would make the record reflect dominant LPAR2 physiology.

- **Well-supported existing MF/CC terms to retain:** GO:0070915 (LPA receptor activity), GO:0004930 (GPCR activity), GO:0030165 (PDZ domain binding), GO:0007204 (positive regulation of cytosolic Ca²⁺).
- **Avoid** collapsing to generic "protein binding."
- **Paralog note (LPAR3/Q9UBY5):** evaluate separately; its GO:0007189 rests on the *same* Sf9 experiment

P 10488122

and the *same* melanocortin-derived MECA IBA, and Im 2000 found no cAMP inhibition + clear Gi coupling — so the identical non-core qualifier applies.

Mechanistic Scope

- **Immediate molecular function tested:** receptor-mediated *increase* in intracellular cAMP (positive coupling to adenylyl cyclase) after LPA binds LPAR2.
- **Direct gene-product activity:** the only positive datum (Bandoh Sf9) is **PTX-insensitive and forskolin-dependent** → indirect **potentiation** of already-active cyclase (plausibly Gq→PKC→AC type II), not autonomous Gs activation.
- **Native direction is inhibitory:** in epithelia LPAR2 *lowers* compartmentalized cAMP (Gi-type;

P 21299497

P 19221439

and drives Gq/G12-13 pathways

P 21134377

P 16904289

- **Separated from this MF:** Ca²⁺ mobilization (Gq), ERK/MAPK, RhoA-dependent migration/invasion, anti-apoptotic TRIP6/NHERF2 complexes

P 19293149

CFTR/NHE3 transport regulation, and tumor phenotypes are pathway/phenotypic consequences — not the AC-activation MF and must not be conflated with it.

Conflicts and Alternatives

1. **Annotation provenance = paralog artifact (primary conflict):** the IBA is inherited from melanocortin receptors/GPR3 (Gs), not from LPA data — the strongest argument that the "activating" call is over-propagated.
2. **Direction conflict:** native LPAR2 *lowers* cAMP (Gi) and uses Gq/G12; the *activation* signal appears only under forskolin priming in Sf9.
3. **Species/system artifact:** Sf9 insect cells have a distinct G-protein/effector milieu; PTX-insensitive cAMP potentiation there may not recapitulate human native coupling.

4. Construct provenance

(P 10729222):

an early ovarian-tumor EDG4 clone had a C-terminal frameshift; the canonical Q9HBW0 C-terminus (...DSTL, intact PDZ motif) is the corrected genomic consensus. Which ORF Bandoh used (human brain cDNA) is not stated in the abstract and should be confirmed in the full text.

Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Exact EDG4 construct in Bando Sf9 assay	Abstracts P 10488122 & 10729222 (full texts not machine-accessible)	If the frameshifted C-terminal variant was used, the cAMP result may not represent canonical LPAR2	Read Methods/sequence in P 10488122 full text; align ORF to Q9HBW0
Gs vs Gq/ PKC→AC-II mechanism	Bando abstract (PTX-insensitive, forskolin-dependent)	Determines whether GO:0007189 (Gs-type) is the correct term vs a modulation/inhibition term	Cholera-toxin/Gs-minus + PKC-inhibitor + AC-isoform reconstitution assays
Human-native cAMP increase	Native studies show cAMP <i>decrease</i> /Gi/Gq (P 21299497 P 19221439 P 21134377)	No human-native assay shows LPAR2 <i>activating</i> cAMP	cAMP biosensor in human cells, LPAR2 knock-in, ± PTX ± Gq inhibitor
GO:0007193 absent	UniProt/QuickGO Q9HBW0	Record over-represents "activation," omits dominant inhibition	Curate Gi/AC-inhibition annotation from P 21299497 / 19221439
P 16203867 — resolved in Iteration 3	Retrieved via Europe PMC; confirms LPA2-mediated Gi inhibition of CFTR (cAMP-lowering) in native intestine	Confirms dominant Gi/AC-inhibiting physiology; supports adding GO:0007193 and treating GO:0007189 as non-core	Already resolved; curator may verify full-text figures for magnitude/EC50

Discriminating Tests

1. **Isoform-defined cAMP biosensor** in a human line expressing canonical Q9HBW0 ([...DSTL](#)), ± PTX, ± Gq inhibitor (YM-254890/FR900359), ± PKC inhibitor, ± forskolin — cleanly separates Gs activation vs Gq/PKC→AC-II potentiation vs Gi inhibition.
2. **Construct-controlled comparison:** canonical vs frameshifted C-terminal EDG4 side-by-side in Sf9 and mammalian cells.
3. **Gα knockout / minigene + BRET** G-protein dissociation panels to read which Gα LPAR2 engages natively.
4. **AC-isoform reconstitution** (AC-II vs AC-V/VI) to test the conditional cyclase-II hypothesis.
5. **Re-audit the PANTHER PTHR22750 node PTN002733616** to confirm the IBA propagates from melanocortin/GPR3 leaves — a curator-actionable check against family-level over-annotation.

Curation Leads (require curator verification)

- **Action:** Reclassify GO:0007189 on Q9HBW0 as **non-core / context-qualified** and **re-ground it on experimental evidence**

([P 10488122](#))

rather than the melanocortin-derived IBA; optionally down-weight/flag the IBA as MECA-family over-propagation. **Add GO:0007193** to represent dominant cAMP-lowering physiology.

- **Candidate reference + exact snippets to verify:**

• ([P 10488122](#)):

"In EDG7- or EDG4-expressing Sf9 cells, LPA stimulated forskolin-induced increase in intracellular cAMP levels, which was not observed in EDG2-expressing cells." / "Neither the EDG7- nor EDG4-transduced Ca(2+) response or cAMP accumulation was inhibited by pertussis toxin."

• ([P 21299497](#))

(for GO:0007193 lead): *"it elevates cAMP levels in proximity to CFTR and upregulates its channel activity"* (upon disrupting LPA2-NHERF2 → intact LPA2 lowers cAMP).

- P 19221439

"NHERF2 confers inhibitory signals by coupling the LPA receptor to CFTR."
- P 21134377

"MAGI-3 increased the interaction of LPA(2) with Ga(12), whereas NHERF-2 preferentially promoted interaction between LPA(2) and Ga(q)."
- P 16203867

(for GO:0007193 lead): *"LPA inhibited CFTR-dependent iodide efflux through LPA2-mediated Gi pathway."*
- **Candidate qualifier note:** activation effect is heterologous (Sf9), forskolin-dependent, PTX-insensitive → conditional Gq/PKC→AC-II potentiation, not canonical Gs.
- **Suggested curator questions:** (1) Which EDG4 ORF did Bando use (canonical vs frameshift)? (2) Should the IBA GO:0007189 be dropped/qualified given its `withFrom` is entirely melanocortin/GPR3? (3) Should GO:0007193 be added? (4) Retrieve
- P 16203867

full text.
- **Suggested experiments:** the isoform-defined cAMP biosensor discriminating assay and PANTHER node audit above.

Limitations

- Full texts of

- P 10488122

10727522 / 16203867 were not machine-accessible in this run; claims rely on complete abstracts plus the seed's Methods excerpt. All quoted snippets are verbatim from retrieved abstracts and should be re-verified against full text.
- Analysis is literature- and database-driven; no wet-lab validation. UniProt/QuickGO reads and the `withFrom` identity resolution are direct programmatic results; the Gq/PKC→AC-II mechanistic attribution is inference, clearly labeled as such.

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